Programming Languages: A Beginner’s Guide

Simple definitions, common uses, and short code examples for 11 popular programming languages.

01 Java

Definition: Java is a high-level, class-based programming language designed to run on many different operating systems through the Java Virtual Machine (JVM).

Common uses: Enterprise applications, Android development, and large backend systems.

```
System.out.println("Hello, world!");
```

02 JavaScript

Definition: JavaScript is a dynamic programming language used to make web pages interactive. It also runs outside browsers, notably on servers using Node.js.

Common uses: Interactive websites, web applications, and backend APIs.

```
console.log("Hello, world!");
```

03 TypeScript

Definition: TypeScript is a programming language built on JavaScript that adds static type checking. TypeScript code is converted into JavaScript to run.

Common uses: Large web applications and projects needing safer, easier-to-maintain JavaScript.

```
const greeting: string = "Hello, world!";
```

04 Python

Definition: Python is a high-level, general-purpose language known for readable syntax and a broad ecosystem of libraries.

Common uses: Automation, data analysis, artificial intelligence, web development, and scripting.

```
print("Hello, world!")
```

05 C++

Definition: C++ is a compiled programming language that extends C with features such as classes, templates, and stronger abstraction tools. It offers detailed control over performance and memory.

Common uses: Game engines, systems software, simulations, and high-performance applications.

```
std::cout << "Hello, world!";
```

06 C#

Definition: C# (pronounced “C sharp”) is a modern, statically typed language developed by Microsoft and commonly used with the .NET platform.

Common uses: Business software, web services, desktop applications, and Unity games.

```
Console.WriteLine("Hello, world!");
```

07 C

Definition: C is a compiled, general-purpose programming language that provides low-level access to memory and hardware-related operations. It has influenced many later languages.

Common uses: Operating systems, embedded devices, firmware, and system utilities.

```
printf("Hello, world!\n");
```

08 Ruby

Definition: Ruby is a dynamic, object-oriented programming language designed to emphasize simplicity and developer productivity.

Common uses: Web applications, especially with Ruby on Rails, and scripting.

```
puts "Hello, world!"
```

09 Go

Definition: Go (also called Golang) is a compiled, statically typed language designed for simplicity, fast builds, and built-in concurrency support.

Common uses: Cloud services, networking tools, backend systems, and infrastructure software.

```
fmt.Println("Hello, world!")
```

10 Rust

Definition: Rust is a compiled systems programming language focused on performance and memory safety without requiring a garbage collector.

Common uses: System tools, performance-sensitive services, embedded systems, and WebAssembly.

```
println!("Hello, world!");
```

11 PHP

Definition: PHP is a server-side scripting language widely used to create dynamic web pages and web applications.

Common uses: Websites, content management systems such as WordPress, and backend services.

```
echo "Hello, world!";
```

Note: Code snippets illustrate the syntax of each language. Some require imports, a class, or a function to run as a complete program.